Myeongsu Seong (成明洙), PhD, FHEA, MIET

Assistant Professor

Citizenship: Republic of Korea
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Address: EB324, Engineering Building, Xi'an Jiaotong-Liverpool University, Suzhou Dushu Lake Science and Education Innovation District, Suzhou Industrial Park, Suzhou 215123, China

EDUCATION

Postgraduate Certificate in Teaching and Supporting Learning in Higher Education (Pass with Merit), University of Liverpool, Liverpool, United Kingdom, 2024.08

Doctor of Philosophy, Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology, Gwangju, Republic of Korea, 2018.08

- Supervisor: Dr. Jae Gwan Kim
- Dissertation: Development of combined diffuse optical and diffuse correlation spectroscopic systems and its application in Alzheimer's disease
- Specialized in biophotonics
- Awarded *Excellent Research Award* by the institute due to academic excellence

Master of Science, Department of Medical System Engineering, Gwangju Institute of Science and Technology, Gwangju, Republic of Korea, 2014.08

- Supervisor: Dr. Jae Gwan Kim
- Thesis: Application of Raman spectroscopy in diagnosis and observation of chemotherapeutic efficacy of breast cancer in small animal
- Specialized in biophotonics

Bachelor of Engineering, Department of Medical Engineering, Jungwon University, Goesan, Chungbuk, Republic of Korea, 2012.08

- Graduated one semester earlier due to academic excellence

RESEARCH EXPERIENCE

Assistant Professor, Department of Mechatronics and Robotics, School of Advanced Technology, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China, 2023.08 – present

Associate Professor (校聘副教授), Research Center for Intelligent Information Technology, Nantong University, Nantong, Jiangsu, China, 2020.12 – 2023.06

Postdoctoral Research Fellow (博士后研究员、助理研究员), University of Michigan-Shanghai Jiao Tong University Joint Institute, Shanghai Jiao Tong University, Shanghai, China, 2018.10 – 2020.09

- Worked as a postdoctoral research fellow at the Optical Imaging Laboratory, University of Michigan-Shanghai Jiao Tong University Joint Institute

- Supervisor: Dr. Sung-Liang Chen
- Gained skills in photoacoustics including relevant hands-on skills in optical and acoustic systems, signal and image processing, and hardware control
- Worked on the development of acoustic-resolution and optical-resolution photoacoustic microscopy, nonlinear photoacoustic microscopy, and deep learning-based optimization of plasmonic simulation

Visiting Scholar, Department of Biomedical Engineering, University of Kentucky, Lexington, Kentucky, US., 2015.07 – 2015.12

- Worked as a visiting scholar at the Biomedical Optics Laboratory, Department of Biomedical Engineering, University of Kentucky
- Supervisor: Prof. Guoqiang Yu
- Contributed to building a low-cost bare charge-coupled device (CCD) camera-based blood flow measurement system. Experienced research involving human subjects, and learned working principles of diffuse correlation spectroscopy, an optical technique to monitor deep tissue blood flow in a non-invasive manner

Professional Research Personnel, Gwangju Institute of Science and Technology, Gwangju, Republic of Korea, 2015.02 – 2018.02

- Worked as technical research personnel to substitute mandatory military service in Korea, only selected people or people at specific institutions are eligible to serve in this type of military service.

Research Assistant, Gwangju Institute of Science and Technology, Gwangju, Republic of Korea, 2012.09 – 2018.08

- Worked as a research assistant during master's and Ph.D. courses at Biophotonics Laboratory, Department of Biomedical Science and Engineering (and Medical System Engineering at the previous time), Gwangju Institute of Science and Technology.
- Techniques including optical systems, electronic systems, animal experiments, signal processing, simple machine learning, and programming were acquired and used.
- Developed various low-cost diffuse optical/speckle/correlation spectroscopic techniques for deep tissue blood flow/oxygenation measurements and performed animal experiments including breast cancer and Alzheimer’s animal models.

TEACHING EXPERIENCE

Lecturing-related experiences

Engineering Drawing (MEC106, co-teaching), AY2425-S1; AY2526-S1

Experimental, Computer Skills and Sustainability (MEC104, co-teaching), AY2324-S2; AY2425-S2 (module leader); AY2526-S2

Entrepreneurial Skills for Advanced Technologies (SAT102, co-teaching), AY2526-S2

Research-led and Project-based Learning (LIF001, mentor), AY2526-S2

Foundations of Computer Science and Engineering (SAT006, co-teaching), AY2324-S1

医学信息学英语 (**English in Biomedical Informatics**, undergraduate course, 48 hours), Department of Biomedical Informatics, School of Medicine, Nantong University, Nantong, China, 2023.02 – 2023.04 (participates as one of the instructors)

- This course covers various topics in bioinformatics including electronic health records, telemedicine, medical information retrieval, article translation, and abstract writing. I teach 24 hours (half) of the course.

Mentoring-related experiences

Supervising two PhD students, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China, 2024.09 – present

Supervised master's theses of three MSc students, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China, 2023.09 – 2025.04

Supervising bachelor's theses of six undergraduate students and supervised 10 undergraduate students' theses, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China, 2023.09 – present

Part-time lecturer, Dong-in Elementary School, Goesan, Chungbuk, Republic of Korea, 2009 – 2010

- Helped elementary school students from different grades for reviewing subjects (among Korean, Math, and English) in which each student was weak.

GRANTS

- Collaborative research project (Ruijin–XJTLU Intelligent Medicine Institute), Medical-Engineering Interdisciplinary Research Grant, 2026.07 – 2027.06, 50,000 RMB, **Co-principal investigator**
- University internal project (XJTLU), FOSA2506035, Postgraduate research fellowship (PGRS), awarded in 2025.07, 2026.03 – 2029.02, 297,000 RMB, 3-year tuition fee of a PhD student, **Principal investigator**
- University internal project (XJTLU), SURF-2025-0143 and SURF-2025-0144, Summer Undergraduate Research Fellowship (SURF, supervised four undergraduate students), 10 weeks, 2025.06 – 2025.08, 6,000 RMB, completed, **Principal investigator**
- University internal project (XJTLU), FOSA2406037, Postgraduate research fellowship (PGRS), 2024.09 – 2027.08, 297,000 RMB, 3-year tuition fee of a PhD student, **Principal investigator**
- University internal project (XJTLU), SURF-2024-0378, Summer Undergraduate Research Fellowship (SURF, supervised four undergraduate students), 10 weeks, 2024.06 – 2024.08, 6,000 RMB, completed, **Principal investigator**
- University internal project (XJTLU), RDF-23-01-119, Research Development Fund, 36 months, 2024.01 - 2026.12, 100,000 RMB, ongoing, **Principal investigator**
- Provincial grant, BK20220603, Jiangsu Young Scientist Fund (江苏省基础研究计划自然科学基金-青年基金项目), 36 months, 2022.07 – 2025.06, 200,000 RMB, completed, **Principal investigator**
- National grant, NRF-2015H1A2A1032268, Development of an optical hemodynamic measurement system and monitoring acute and chronic brain diseases using the developed system (Global Ph.D. Fellowship), 36 months, 2015.03 – 2018.02 (worked on the project: 2015.03 – 2018.02), approximately 540,000 RMB (90,000,000 Korean Won), completed, **Principal investigator**

AWARDS / DISTINCTIONS

- Jiangsu Foreign Expert Workshop (江苏省外国专家工作室, provincial-level platform project (省级科研平台项目), **secured it as the leader (领衔外国专家)** of foreign scholars at the Research Center for Intelligent Information Technology, Nantong University), started from 2022.12 and officially announced in 2023.02, left Nantong University in 2023.06
- Young Foreign Talent Plan of National Foreign Expert Project (国家外国专家项目中"外国青年人才计划"), Ministry of Science and Technology (MOST) of China (中国科学技术部), supported till 2022.12
- Excellent Research Presentation Award (as a second author), Optical Society of Korea Summer Meeting, Republic of Korea, 2018.08
- Excellent Research Award (**awarded for research excellence during the Ph.D. study, only one student in the department**), Gwangju Institute of Science and Technology, Republic of Korea, 2018.08
- Best Presentation Award, The 3rd IEEE EMBS International Summer School of Neural Engineering held by the Department of Biomedical Engineering, Shanghai Jiao Tong University, China, 2017.08
- IT Innovative Idea Award (2016 Qualcomm-GIST Innovation Award), School of Electrical Engineering and Computer Science, Gwangju Institute of Science and Technology (Supported by Qualcomm Korea), Republic of Korea, 2016.12
- iMSE (Institute of Medical System Engineering) Scholarship (**Only three students in the department based on academic performance**), Department of Biomedical Science and Engineering, Gwangju Institute of Science and Technology, Republic of Korea, 2016.12
- Global Ph.D. Fellowship (**One of the most prestigious fellowships for Ph.D. students with high potential in Korea, only applicable once in a lifetime**), KRW 90,000,000 (around 540,000 RMB), National Research Foundation of Korea, Republic of Korea, 2015.03 – 2018.02
- Outstanding Research Award (Conference Presentation), The Korean Society of Medical and Biological Engineering, Republic of Korea, 2014.05
- Best Research Award (as a second author), Optical Society of Korea Summer Meeting, Republic of Korea, 2013.07
- Government Supported Full Tuition Fee Waiver, Gwangju Institute of Science and Technology, Republic of Korea, 2012.09 – 2018.08
- Scholarships Awarded Six Times for Excellent Academic Achievements, Jungwon University, Republic of Korea, 2009.03 – 2012.08

PROFESSIONAL MEMBERSHIPS / ORGANIZATIONS / SERVICE

- Member (MIET), Institution of Engineering and Technology, 2025 – present
- Member, IEEE, 2025 – present
- Fellow of Advance HE (FHEA), Advance HE (higher education), 2024 – present
- Member, Optical Society of Korea, 2024 – present
- Reviewer for journals: APL Photonics, IEEE Transactions on Instumentation and Measurement, Applied Physics Letters, Computers in Biology and Medicine, IEEE Access, IEEE Open Journal of Engineering in Medicine and Biology, Journal of Physics: Photonics, Journal of Biomedical Optics,

Journal of Biophotonics, Neurophotonics, Optics Letters, Scientific Reports, BMC Research Notes, AIP Advances, Biosensors, Photonics, Sensors, Spectroscopy Journal, and so forth

- Technical program committee member for conferences: 10th Asia Conference on Mechanical Engineering and Aerospace Engineering (October 18, 2024 – October 20, 2024); International Conference on Smart Computing, IoT and Machine Learning (July 25, 2024 – July 27, 2024) and more
- Reviewer for conferences: 2025 4th International Conference On Multidisciplinary Applications of Information Technology (reviewed 1 manuscript, August 25, 2025 – August 26, 2025); International Conference on Smart Computing, IoT and Machine Learning 2025 (reviewed 1 manuscript, June 3, 2025 – June 4, 2025); 10th Asia Conference on Mechanical Engineering and Aerospace Engineering (reviewed 6 manuscripts, October 18, 2024 – 20 October, 2024); International Conference on Smart Computing, IoT and Machine Learning 2024 (reviewed 5 manuscripts, June 6, 2024 – June 7, 2024); 10th International Conference on Smart Computing and Communication (reviewed 2 manuscripts, July 25, 2024 – July 27, 2024) and more
- External examiner for Ph.D. dissertation: Youngjoo Lee at GIST (November 21, 2023); Minhee Kim at GIST (November 18, 2024); Yoonho Oh at GIST (November 27, 2024); Nourelhuda Ali Yousif Mohamed at GIST (May 19, 2025); Manal Mustafa Mohamedali Mohamed at GIST (May 19, 2025); Semere Araya Asefa at Yonsei University Mirae Campus (June 9, 2025); Dong-Hyuk Choi at GIST (November 26, 2025); Sangmin Shim at Yonsei University Mirae Campus (December 2, 2025)
- Department Student Representative, Department of Medical System Engineering, Gwangju Institute of Science and Technology, 2015.03 – 2016.03

PUBLICATIONS

Peer-reviewed journal articles

- Yoonho Oh[*], **Myeongsu Seong**[*], Sungchul Kim, Seonghyun Kim, Jaeyoung Bae, Jae Gwan Kim[#], Jae Yoon Hwang[#] "Anesthesia monitoring (tentative title)," (in preparation)
- Jiachen Zhou, Junyu Yao, Anh Nguyen, Jeong Hyeon Park, Lei Fu, Jinxin Gu[#], and **Myeongsu Seong**[#] "C. elegans quantification (tentative title)," (in submission)
- Muhammad Jawad Hussain[#], Heming Bai, **Myeongsu Seong**, and Shahbaz Hassan Wasti "Integrating lexical semantics for deep conceptual similarity: A WordNet-based vector framework," Information Sciences 755, 123757, 2026
- Siqi Liang, Boxiang Liu, Chaoyang Chen, Mingyang Lv, Yue Hu, Qingshan Liu, Sung-Liang Chen[#], and **Myeongsu Seong**[#] "Restoration of out-of-focus region for photoacoustic endoscopic imaging enabled by convolutional neural networks," Physica Scripta 101, 216005, 2026
- Jiachen Zhou, Siqi Liang[#], Jeong Hyeon Park, Anh Nguyen, and **Myeongsu Seong**[#] "Burn depth assessment by photoacoustic imaging: a review," Methods 252, 65-82 2026
- Tianru Zhang[#], Huangjia Wu[*], Yixin Feng[*], Quan Zhang[*], Myeongsu Seong[*], Jie Sun[*], Mark Leach[*], and Eng Gee Lim[*] "Construction of an educational teaching evaluation system under the context of deep integration of artificial intelligence and teaching practices," Acta Pedagogia Asiana 5(SI), 45-54, 2026
- Seokgyu Kwon, Sangmin Shim[*], Kyunghyun Yu, **Myeongsu Seong**[#], and Dasol Lee[#] "Advances in electrospun nanofibers for biomedical engineering," Fibers and Polymers 27, 565-599, 2026 2025
- Siqi Liang, Nan Wan[*], Guo Chen[*], Sung-Liang Chen[#], and **Myeongsu Seong**[#] "Sensing changes in triglyceride concentration in blood solution using diffuse optical spectroscopy," Optics and Laser Technology 197, 114767, 2026
- Minseo Jeong, Seokgyu Kwon, Changhwan Hyeon, Juhoon Baek, **Myeongsu Seong**, Minkyung Kim[#], and Dasol Lee[#] "Hydrophobic radiative cooling using zein-functionalized polyvinyl alcohol nanofibers with dielectric nanoparticles," Small 21(45) e07295, 2025

- Hyeryun Jeong, **Myeongsu Seong**#, Kwangsung Park, and Jae Gwan Kim# "Optical assessment of genital changes associated with female sexual arousal: A mini-review," Sexual Medicine 13(5) qfaf065, 2025
- Siqi Liang, Nan Wan*, Wenxuan Pi*, Ke Zhang, Jingfei Wen, and Sung-Liang Chen#, **Myeongsu Seong**# "Photoacoustic microscopy for visualization of stents in multiple scenarios," Optics Letters 50(15) 4722-4725, 2025
- Changhwan Hyeon, Minseo Jeong*, Seokgyu Kwon, Juhoon Baek, **Myeongsu Seong**, Minkyung Kim, and Dasol Lee# "Thermal protection of wearable devices under outdoor conditions using radiative cooling films," Optical Materials 167, 117258, 2025
- Muhammad Jawad Hussain#, **Myeongsu Seong**, Behjat Shahid, and Heming Bai# "Assessing PM2.5 exposures across the Yangtze River Delta region by filling satellite aerosol optical depth and top-of-atmosphere retrieval gaps using a temporal and spatial convolutional approach," Toxics 13(5), 392, 2025
- Jingfei Wen, **Myeongsu Seong**, Guo Chen, and Sung-Liang Chen# "Low-cost diffuse optical spectroscopy for assisting sclerotherapy: A proof-of-concept study," Optics and Laser Technology 186, 112664, 2025
- Semere Asefa, **Myeongsu Seong**#, and Dasol Lee# "Design of Bilayer Crescent Chiral Metasurfaces for Enhanced Chiroptical Response," Sensors 25(3), 915, 2025
- Zhe Li, Hua Zhang#, Lei Sheng#, Kaifei Nong, Kailong Wang, Zilong Wang, Zhendong Zhang, **Myeongsu Seong** "Liquid-immersed thermal management to cylindrical lithium-ion batteries for their pack applications," Journal of Energy Storage 85, 111060, 2024
- Semere Asefa, Sangmin Shim, **Myeongsu Seong**#, Dasol Lee# "Chiral metasurfaces: a review of the fundamentals and research advances," Applied Sciences 13(19) 10590, 2023
- **Myeongsu Seong**# "Comparison of numerical-integration-based methods for blood flow estimation in diffuse correlation spectroscopy," Computer Methods and Programs in Biomedicine 241, 107766, 2023
- Nan Wan, Zhe Li, **Myeongsu Seong**, Wei Niu, Rong Wu, and Sung-Liang Chen# "Sensing of triglyceride concentration in blood solution using photoacoustic microscopy," Optics Letters 48(14), 3769-3772, 2023
- Nan Wan, Pengcheng Zhang*, Zuheng Liu, Zhe Li, Wei Niu, Xiuye Rui, Shibo Wang, **Myeongsu Seong**, Pengbo He, Siqi Liang, Jiasheng Zhou, Rui Yang#, Sung-Liang Chen# "Implantable QR code subcutaneous microchip using photoacoustic and ultrasound microscopy for secure and convenient individual identification and authentication," Photoacoustics 31, 100504, 2023
- **Myeongsu Seong**, Yoonho Oh, Hyung Joon Park, Won-Seok Choi#, and Jae Gwan Kim# "Use of hypoxic respiratory challenge for differentiating Alzheimer's disease and wild-type mice non-invasively: A diffuse optical spectroscopy study," Biosensors 12(11), 1019, 2022
- **Myeongsu Seong**, Yoonho Oh*, Kijoon Lee#, and Jae Gwan Kim# "Blood flow estimation via numerical integration of temporal autocorrelation function in diffuse correlation spectroscopy," Computer Methods and Programs in Biomedicine 222, 106933, 2022
- Heming Bai#, Yuli Shi, **Myeongsu Seong**, Wenkang Gao, and Yuanhui Li "Influence of spatial resolution on satellite-based PM2.5 estimation: implications for health assessment." Remote Sensing 14(12), 2933, 2022
- Heming Bai#, Rusha Yan, Wenkang Gao, Jing Wei, and **Myeongsu Seong** "Spatial representativeness of PM2.5 monitoring stations and its implication for health assessment," Air Quality, Atmosphere & Health 15, 1571-1581, 2022
- **Myeongsu Seong**, Wenzhao Yang, Yujie Han, Jiasheng Zhou, Lili Jing, and Sung-Liang Chen# "Investigation of nonlinear photoacoustic microscopy using a low-cost infrared lamp," Journal of Biophotonics 15(4), e20210301222, 2022
- Wenzhao Yang, Jiasheng Zhou, Weihao Shao, **Myeongsu Seong**, Pengbo He, Zhanhong Ye, Zhendong Guo, Lili Jing, and Sung-Liang Chen# "Photoacoustic-fluorescence microendoscopy *in vivo*." Optics Letters 46(10), 2340-2343, 2021

- Hyeryun Jeong, Hyun-Suk Lee, **Myeongsu Seong**, Jaewoo Baek, Kwangsung Park, and Jae Gwan Kim[#] "Changes of apomorphine-induced vaginal hemodynamics in an ovariectomized rat model using near-infrared spectroscopic probe." The Journal of Sexual Medicine 18(8), 1328-1336, 2021
- Nan Wan, **Myeongsu Seong**, and Sung-Liang Chen[#] "Theoretical Investigation of photoacoustics from cancer cells: modified models" IEEE Journal of Selected Topics in Quantum Electronics 27(5), 1-10, 2021
- Heming Bai, Lei Jiang*, Ting Li*, …, **Myeongsu Seong**,…, Huji Xu[#] "Acute effects of air pollution on lupus nephritis in patients with systemic lupus erythematosus: A multicenter panel study in China," Environmental Research 195, 110875, 2021
- **Myeongsu Seong** and Sung-Liang Chen[#] "Recent advances toward clinical applications of photoacoustic microscopy: a review." Science China: Life Sciences 63, 1-15, 2020
- Hyeryun Jeong, **Myeongsu Seong**, Kwangsung Park, and Jae Gwan Kim[#] "Monitoring differences of vaginal hemodynamic and temperature response for sexual arousal by different anesthetic agents using an optical probe." Current Optics and Photonics 4(1), 57-62, 2020
- Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, Sucbei Moon, and Jae G. Kim[#] "Design of an optical probe to monitor vaginal hemodynamics during sexual arousal." Sensors 19(9), 2129, 2019
- **Myeongsu Seong**, Phuong Minh Mai, Kijoon Lee, and Jae G. Kim[#] "Simultaneous blood flow and oxygenation measurements using an off-the-shelf spectrometer." Chinese Optics Letters 16(7), 071701, 2018
- Songhyun Lee, Hyeryun Jeong, **Myeongsu Seong**, and Jae G. Kim[#] "Change of tumor vascular reactivity during tumor growth and postchemotherapy observed by near-infrared spectroscopy." Journal of Biomedical Optics 22(12), 121603, 2017
- **Myeongsu Seong**, NoSoung Myoung*, Songhyun Lee, Hyeryun Jeong, Sang-Youp Yim[#], and Jae G. Kim[#] "Longitudinal Raman spectroscopic observation of skin biochemical changes due to chemotherapeutic treatment for breast cancer in small animal model." Journal of Spectroscopy 2017(4), 1-9, 2017
- Chong Huang, **Myeongsu Seong**, Joshua Paul Morgen, Siavash Mazdeyasna, Jae G. Kim, Jeffrey Todd Hastings, and Guoqiang Yu[#] "Low-cost compact diffuse speckle contrast flowmeter using small laser diode and bare charge-coupled-device." Journal of Biomedical Optics (Letters) 21(8), 080501, 2016
- **Myeongsu Seong**, Zephaniah Phillips V*, Phuong Minh Mai, Chaebeom Yeo, Cheol Song, Kijoon Lee, and Jae G. Kim[#] "Simultaneous blood flow and oxygenation measurements using a combination of diffuse speckle contrast analysis and near-infrared spectroscopy." Journal of Biomedical Optics 21(2), 027001, 2016

*: co-first; [#]: corresponding author(s)

EI-indexed conference proceedings

- Jingxuan Qian and **Myeongsu Seong**[#] "YOLO-ASFF: A model for detecting tomatoes of different maturities based on improved YOLO v5s," Submitted to and accepted by 2025 7th International Conference on Software Engineering and Computer Science
- Tianru Zhang[#], Huangjia Wu, Yixin Feng, Quan Zhang, **Myeongsu Seong**, Jie Sun, Mark Leach, and Eng Gee Lim "Construction of an educational teaching evaluation system under the context

of deep integration of artificial intelligence and teaching practices," Submitted to and accepted by the International Conference on Higher Education Learning and Teaching 2024

- **Myeongsu Seong***, Zephaniah Phillips*, Phuong Minh Mai, Chaebeom Yeo, Cheol Song, Kijoon Lee, and Jae G. Kim "A fiber optic probe coupled low-cost CMOS camera-based system for simultaneous measurement of oxy-, deoxyhemoglobin, and blood flow." Proceedings of SPIE, In International Conference on Nano-Bio Sensing, Imaging, and Spectroscopy 2015, vol. 9523, p. 95230E. International Society for Optics and Photonics, 2015. *: co-first.
- Songhyun Lee, **Myeongsu Seong**, Hyeryun Jeong, and Jae G. Kim "Tumor vascular reactivity as a marker to predict tumor response to chemotherapy." Proceedings of SPIE, In Optical Tomography and Spectroscopy of Tissue XI, vol. 9319, p. 93190E. International Society for Optics and Photonics, 2015.
- **Myeongsu Seong**, Myoung NoSoung, Sang-Youp Yim, and Jae G. Kim "Longitudinal in vivo transcutaneous observation of Raman signals from breast cancer during chemotherapy in small animal model." Proceeding of SPIE, In Photonic Therapeutics and Diagnostics XI, vol. 9303, p.93032U. International Society for Optics and Photonics, 2015.

*: co-first; #: corresponding author(s)

PATENT

- Jae Gwan Kim, Songhyun Lee, Hoonsup Kim, **Myeongsu Seong**, and Kwangsung Park, Device for diagnosis and treatment of erectile dysfunction. Country: Republic of Korea. Since 2016.02.25

INVITED TALKS

- **Myeongsu Seong** "Near-Infrared Spectroscopy for Biomedical Applications", Date: February 20, 2026 (period: February 18, 2026 - February 21, 2026). Organizer: Institute of Electronics and Information Engineers (IEIE) (International Conference on Electronics, Information, and Communication (ICEIC) 2026)
- **Myeongsu Seong** "Photoacoustic Microscopy for Visualization of Stents in Multiple Scenarios", Date: August 26, 2025. Invited by Prof. Dasol Lee. Organizer: Bio-Nanophotonics System Laboratory, Department of Biomedical Engineering, Yonsei University, Korea
- **Myeongsu Seong** "A brief introduction of optical tools for biomedical engineering researches", Date: May 24, 2025. 2025 Spring Conference of Korean Scientists and Engineers Association in China
- **Myeongsu Seong** "의학과 공학의 만남이 열어가는 길: 의공학 (A path opened by the meeting of medicine and engineering: biomedical engineering)", Date: May 23, 2025. Shanghai Korean School (for high school students who will enter a college soon)
- **Myeongsu Seong** "Classification of Alzheimer's and age-matched animals using hemodynamic signals during hypoxic gas challenge", Date: November 7, 2024 (period: November 7, 2024 - November 9, 2024). Organizer: International Biomedical Engineering Conference 2024 (jointly held with 2024 Autumn Conference of the Korean Society of Medical and Biological Engineering)
- **Myeongsu Seong** "Meat Freshness Monitoring Using Speckle Imaging", Date: August 26, 2024. Invited by Prof. Dasol Lee. Organizer: Department of Biomedical Engineering, Yonsei University, Korea

- **Myeongsu Seong** “Advancements in deep tissue optical blood flow monitoring technology”, Date: November 10, 2023 (period: November 9, 2023 - November 11, 2023). Organizer: International Biomedical Engineering Conference 2023 (jointly held with 2023 Autumn Conference of the Korean Society of Medical and Biological Engineering)
- **Myeongsu Seong** “Diffuse-optics based flow measurement: towards simplified systems”, Date: November 16, 2021. Invited by Prof. Dasol Lee. Organizer: Department of Biomedical Engineering, Yonsei University, Korea
- **Myeongsu Seong** “Spatial frequency domain imaging”, Date: January 31, 2023. Invited by Prof. Dasol Lee. Organizer: Bio-Nanophotonics System Laboratory, Department of Biomedical Engineering, Yonsei University, Korea

PRESENTATIONS

Oral presentations

- Jingxuan Qian and **Myeongsu Seong**# “YOLO-ASFF: A model for detecting tomatoes of different maturities based on improved YOLO v5s,” Submitted to and accepted by 2025 7th International Conference on Software Engineering and Computer Science
- SPIE Photonics West. Yoonho Oh, **Myeongsu Seong**, Sungchul Kim, Sunghyun Kim, and Jae G. Kim “Optimization of DRS-DCS system for measurement of tissue metabolism.” Organizer: International Society for Optics and Photonics, 2020
- The Conference of the Optical Society of Korea Summer Session. Yoonho Oh, **Myeongsu Seong**, and Jae G. Kim “Simultaneous measurement of cerebral blood flow and oxygenation depending on depths of isoflurane anesthesia in a rat model.” Organizer: Optical Society of Korea, 2018
- The Conference of the Optical Society of Korea Summer Session. Yoonho Oh, **Myeongsu Seong**, and Jae G. Kim “Cerebral hemodynamic measurement by software correlator based diffuse correlation spectroscopy.” Organizer: Optical Society of Korea, 2017
- The Conference of the Optical Society of Korea Summer Session. Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, Sucbei Moon, and Jae G. Kim “Differential response of vaginal hemodynamics in a rat model depending on anesthetic agents.” Organizer: Optical Society of Korea, 2017
- Global Ph.D. Fellowship Conference. **Myeongsu Seong**, Phuong Minh Mai, Kijoon Lee, and Jae G. Kim “The simultaneous measurement of blood oxygenation and blood flow based on a commercial spectrometer and two lasers system.” Organizer: National Research Foundation of Korea, 2016
- Global Ph.D. Fellowship Conference. **Myeongsu Seong**, Zephaniah Phillips V, Phuong Minh Mai, Chaebeom Yeo, Cheol Song, Kijoon Lee, and Jae G. Kim. “Simultaneous blood flow and blood oxygenation measurements using a combination of diffuse speckle contrast analysis and near-infrared spectroscopy.” Organizer: National Research Foundation of Korea, 2015
- International Conference on SPIE 2015 Nano-Bio Sensing, Imaging, and Spectroscopy. **Myeongsu Seong***, Zephaniah Phillips V*, Phuong Minh Mai, Chaebeom Yeo, Cheol Song, Kijoon Lee, and Jae G. Kim “A fiber-optic probe coupled low-cost CMOS camera-based system for simultaneous measurement of oxy-, deoxyhemoglobin, and blood flow.” Organizer: International Society for Optics and Photonics, 2015 *: co-first
- SPIE Photonics West. Songhyun Lee, **Myeongsu Seong**, Hyeryun Jeong, and Jae G. Kim “Tumor vascular reactivity as a marker to predict tumor response to chemotherapy.” Organizer: International Society for Optics and Photonics, 2015

- The Conference of the Korean Society of Medical and Biological Engineering Spring Session. **Myeongsu Seong**, NoSoung Myoung, Sang-Youp Yim, and Jae G. Kim "In vivo observation of biochemical composition changes from breast tumor caused by chemotherapeutic agent using Raman spectroscopy." Organizer: Korean Society of Medical and Biological Engineering, 2014
- The 3rd Conference of Korean Society of Optoelectronics. **Myeongsu Seong**, Hyeryun Jeong, Songhyun Lee, NoSoung Myoung, Sang-Youp Yim, and Jae G. Kim "Observation of chemotherapeutic effect on breast cancer in vivo by using Raman spectroscopy." Organizer: Korean Society of Optoelectronics, 2013

Poster presentations

- ISSWSH Annual Meeting. Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, and Jae G. Kim "Development of a multi-channel sensor based on optical fibers for measurement of female genital sexual arousal response." Organizer: International Society for the Study of Women's Sexual Health, 2018.
- Annual Biophotonics Conference. Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, Sucbei Moon, and Jae G. Kim "A multichannel probe to study female sexual dysfunction in a rat model." Organizer: Optical Society of Korea, 2017.
- Annual Biophotonics Conference. **Myeongsu Seong**, Yoonho Oh, and Jae G. Kim "Investigation of how much diffuse correlation spectroscopy signal is affected when it is combined with broadband diffuse optical spectroscopy system." Organizer: Optical Society of Korea, 2017.
- The Optical Society of Korea Winter Meeting. Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, Sucbei Moon, and Jae G. Kim "Study on a female sexual dysfunction in a rat model using a multifunctional probe." Organizer: Optical Society of Korea, 2017.
- SPIE Photonics West. Hyeryun Jeong, **Myeongsu Seong**, Hyun-Suk Lee, Kwangsung Park, and Jae G. Kim "Vaginal hemodynamic changes during sexual arousal in a rat model by diffuse optical spectroscopy." Organizer: International Society for Optics and Photonics, 2017.
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